Paula Moukhtar Sobhy

***Computer Science and Artificial Intelligence student* | Aspiring Machine Learning Engineer**

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Phone: +201206394220 **Portfolio**: My Portfolio

Objective

Machine Learning Engineer with a strong foundation in artificial intelligence, search and optimization techniques and software engineering. Experienced in developing machine learning solutions using Python and modern ML frameworks, with experience in backend development and databases.

Education

- Junior Student at Faculty of Computer Science and Artificial Intelligence – Benha University, Benha, Egypt **(Graduation year 2027)**
- El-Maaraf Modern Language school, Nasooh-street, Hadaak El-Zatoon, Cairo

Internships

Manufacturing Commercial Vehicles (MCV) | Software developer (August 2025 – September 2025)

- Worked on backend tasks using C#, .NET, SQL Server, and API integration.
- Improved backend logic, database handling, and API communication.
- Understood industrial data handling and reporting.
- Gained experience in real-world software workflows through guided projects.

Courses & Certifications

Digital Egyptian Pioneers Initiative - Data Science (6 months)

Comprehensive program covering the full data science pipeline using Python.

- **Topics:** Data handling, cleaning, analysis, visualization, preprocessing, machine learning models, databases, web scrapping and MLOps tools.

Huawei Certified ICT Associate (HCIA–AI) (80 Hours)

- Covered AI concepts, ML/DL models, NN applications and TensorFlow framework.
- ML models (LR, DT, NB, KNN, SVM, RF) and DL models (MLP, Sequential)
- Feature engineering, model training, and hyperparameter tuning.
- Direct practice in data preprocessing, model selection, training, and evaluation.

NTI-ITIDA Cloud Architect Initiative (120 Hours)

Intensive course by Ministry of Communications and Information Technology (MCIT), Egypt

- Topics: cloud fundamentals, infrastructure design, cloud application planning with actual deployment and monitoring of cloud services.

Technical Skills

- **Languages:** Python, SQL, C#, and C++
- **ML & AI:** ML, DL, NLP, CNN, Data Preprocessing, Feature Engineering, Model Evaluation
- **Visualization:** Matplotlib, Seaborn, Plotly
- **Web & Backend:** HTML, CSS, JS, Bootstrap, OOP, .Net, MVC, APIs
- **Technical Writing:** Create clear, concise documentation, reports, and presentations
- **Databases:** MySQL, Microsoft SQL Server
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Streamlit
- **Concepts:** Search & Optimization Techniques
- **Tools:** Git, Jupyter Notebook, VS Code, Anaconda, and Postman

Top Projects

Plant Disease Classification using CNN & MobileNetV2 (GitHub Repo)
Technologies: Python, TensorFlow/Keras, CNN, MobileNetV2, Streamlit

- Developed a deep learning pipeline for plant disease classification across 38 classes using CNNs and transfer learning with MobileNetV2.
- Trained and evaluated multiple deep learning architectures, achieving 95.05% test accuracy with MobileNetV2.
- Implemented preprocessing, training, evaluation, and deployment workflows for an end-to-end machine learning solution.
- Deployed a Streamlit application for real-time inference, prediction visualization, and model comparison.
- Implemented batch image analysis and confidence-based reporting to improve usability and model interpretability.

Network Intrusion Detection System by AI - DEPI Project (GitHub Repo)
Technologies: Python, Scikit-learn, Pandas, Streamlit, Machine Learning

- Network Intrusion Detection System to detect malicious network traffic.
- Random Forest classification model capable of detecting multiple cyber-attack types.
- Evaluated using accuracy, precision, recall, and F1-score.
- Built an interactive Streamlit dashboard for model usage and prediction.
- Implemented a user interface that allows uploading network data and monitoring intrusion detection results.

Connect 4 Game with AI Opponent (GitHub Repo)
Technologies: Python, Pygame, Minimax Algorithm, Alpha-Beta Pruning

- Implemented Player vs Player, Player vs AI, and AI vs AI game modes.
- Designed an AI opponent using the Minimax algorithm with Alpha-Beta pruning to optimize decision-making and reduce computation time.
- Implemented game logic including board representation, move validation, and win detection (horizontal, vertical, and diagonal).

Hotel management system (full stack Project) (GitHub Repo)
- A Web app designed to simplify and automate hotel booking experience.
- Implemented room and booking management with real-time availability checks.
- Designed a secure authentication and authorization system for admins and customers.
- Built using HTML, CSS, JS, Bootstrap, C# with SQL Server and tested with postman.
- Strengthened skills in database design, backend logic, and full-stack development.

Activities & Societies

- **IEEE** - Flutter Team Member.
- **FCAI ACPC Benha Club** – Active member.
- **Nasa Space Apps** - HR Member.
- **AKS** – Scouting Captain.
- **El-Fit Championship –** Operations & Logistic Team Volunteer.

Languages

- **Arabic:** Native
- **English:** Fluent